

# Digital Design Coding Challenge

## Sports Analyst Track | Facilitator Guide

*MindFuel Foundation | Grades 6 to 12 | Two 2.5 hour sessions | Chromebook friendly, browser only, nothing to install*

### What this is

A hackathon rebuilt for students who have never written a line of code. Instead of one big problem, students work through seven small challenges that each award points. Easy challenges are worth 1 to 3 points and anyone can do them. Hard ones are worth 7 to 10 and give experienced kids something to chase. The points do the judging, so there is no judging panel.

Students play the role of a data analyst hired by a basketball team. The team has money for one new player. Across the seven challenges, students learn to read data, rank it, filter it, chart it, and finally make a signing recommendation backed by evidence. Every skill maps to a real job.

### Why it works with zero coding experience

- Nothing to install. The notebooks run in the browser on any Chromebook.
- Students never write code from scratch. They run prepared cells, answer questions, and change single values in working code.
- Scout, a friendly bot inside the notebook, gives hints, checks answers, celebrates wins, and never lets anyone stay stuck. Wrong answers just loop with a nudge.
- Answers are checked by meaning, not exact typing. Spelling "sky" instead of "Sky" still counts. Nobody fails on a quote mark.
- Low floor, high ceiling. A total beginner banks small wins all day. A kid who codes already can sprint to the capstone.

### The event at a glance

#### Session 1: First day on the job (2.5 hours)

- Welcome and framing speech, 10 min
- Challenge 0: Meet the database and make a first chart, 20 min, 2 pts
- Challenge 1: Read the scouting report, 25 min, 3 pts
- Challenge 2: Find the scorers by sorting, 30 min, 3 pts
- Challenge 3: The money question, filtering with two conditions, 40 min, 5 pts

- Scoreboard checkpoint and wrap, 15 min. Buffer built into every block.

## Session 2: The big project (2.5 hours)

- Recap and restart, 10 min
- Challenge 4: Charts that convince the coach, 30 min, 5 pts
- Challenge 5: Your own investigation, open ended, 45 min, 7 pts
- Challenge 6: The pitch, 30 min, 10 pts
- Sharing circle: every student delivers their one sentence pitch out loud, 20 min
- Scoreboard, prizes if using them, close, 15 min

## Before the kids arrive

- Each student needs a Chromebook or any laptop with a browser. That is the whole tech list.
- Open the challenge link yourself and run challenge-0 top to bottom once, so you have seen every screen the kids will see.
- Have the point tracker ready: whiteboard, sticky notes, or a shared slide. Students report their score at the checkpoints.
- Optional prizes: Callysto ran junior and senior streams with gift cards. A made up trophy works too. The points are the real motivator.

## The framing speech

The first ten minutes set the whole tone. You are not introducing a coding class. You are hiring a room full of analysts.

**Say it like this:** *"Congratulations, you are all hired. You now work for a professional basketball team. Here is the situation: the team has budget to sign exactly one new player, and the general manager does not know who to pick. By the end of today, each of you is going to tell the GM who to sign, and you are going to prove it with data. Your assistant is called Scout. Scout lives in your browser, keeps your score, and will help you whenever you are stuck. Easy tasks earn small points, hard tasks earn big points. Do them in order or jump ahead, your choice. Let us get you into the office."*

Then show them how to open the first notebook and how Shift plus Enter runs a cell. That is the only instruction they need to start.

## How to explain the big ideas with no jargon

Use these when a student asks what something is. Each one lands without any computer science background.

- A dataset: "It is a team roster sheet. Every row is a player, every column is a fact about them. That is all data is, facts in rows."
- Running a cell: "The instructions are already written. Shift plus Enter means: do it."
- Sorting: "Line the players up from best to worst at one thing. Tallest to shortest, but for stats."
- Filtering: "Cross out everyone who does not fit. The GM says under 20 million and over 18 points, so everyone else leaves the room."
- A chart: "The same table, but drawn, so your eyes can spot the answer faster than reading numbers."
- The capstone: "Every job interview ever: make a decision and defend it with evidence."

## Challenge by challenge

What happens in each notebook, what to say when introducing it, and where kids get stuck.

### Challenge 0: Welcome to the front office (2 pts)

Students type their name, meet Scout, see the full database, and make a deliberately messy chart of everything at once.

**Say it like this:** *"First job: meet your team. Run the first cell and Scout will introduce you to the database. Then make your first chart. It will look terrible. That is the point, by the end of today your charts will actually mean something."*

- Stuck point: forgetting Shift plus Enter. Fix it once out loud for the room.
- The messy chart is intentional. If a student says "this chart is bad," say "exactly, your job today is to learn to make good ones."

### Challenge 1: Read the scouting report (3 pts)

Multiple choice questions about the table itself. Which player is in the first row, which column means scoring. Pure table reading, zero code.

**Say it like this:** *"Analysts read tables the way you read a game screen: fast. Two questions, no code, just look carefully."*

- Stuck point: kids overthink it. Remind them the answer is literally on screen.

## Challenge 2: Find the scorers (3 pts)

Scout sorts the table by points per game and asks who leads. Then students change one word in a working code cell, the column name, to answer a new question. This is their first code edit.

**Say it like this:** *"The GM texted: who are our best scorers? Sorting answers ranking questions in one line. Then you get to change that line, one word, and suddenly it answers a completely different question. That is what coding actually is."*

- Stuck point: typing the column name wrong. Scout lists every valid column if they get it wrong, so point them at the hint.

## Challenge 3: The money question (5 pts)

The core analyst skill: filtering with two conditions at once. At least 18 points per game AND salary under 20 million. Students read the filtered result and answer who fits.

**Say it like this:** *"Anyone can want the best player. The job is finding the best player you can afford. Two conditions at the same time. Watch how fast the computer crosses out everyone who does not fit."*

- Stuck point: none usually. This one lands. End Session 1 here with the scoreboard cell.

## Challenge 4: Charts that convince the coach (5 pts)

Session 2 opener. A scatter plot of salary against scoring with every player labelled. Students learn to read the top left corner, high scoring and cheap, as the value zone.

**Say it like this:** *"Numbers convince analysts. Charts convince coaches. This chart has a magic corner: top left is players who score a lot and cost a little. Smart teams live there. The movie Moneyball is about this exact corner."*

- Stuck point: reading a scatter plot is new for many. Trace one player with your finger: across for cost, up for scoring.

## Challenge 5: Your own investigation (7 pts)

Open ended. Three working code cells, sort, filter, chart, sit under the challenge. Students copy, modify, and chase any question they choose, then write one sentence about what they found.

**Say it like this:** *"You have every skill now. So here is the deal: you choose the question. Best passer for the money? Do centers really rebound more? Pick something you actually want to know, change the code, find out. There is no wrong question."*

- Stuck point: freedom freezes some kids. Give them the menu from the notebook: passers per dollar, positions and rebounds, games played. Choosing from three is easier than choosing from infinity.
- Fast kids: challenge them to combine three conditions in one filter, or chart two things nobody else picked.

## Challenge 6: The pitch (10 pts)

The finale. Scout asks who the team should sign. Students name a player, Scout pulls that player's full stat line, and then asks for the why. A one word reason gets bounced, Scout requires an actual sentence of evidence.

**Say it like this:** *"The GM just walked in. One sentence: who do we sign and why. Scout will not accept "because he is good." Neither would a real GM. Back it up with the numbers you found today."*

- Stuck point: none. This is the payoff. Everyone finishes with a recommendation.

## The sharing circle

After challenge 6, go around the room. Each student or team stands and delivers the one sentence pitch out loud. Thirty seconds each. This is the demo day moment of a real hackathon, shrunk to a sentence so nobody needs slides or stage time. Applaud every single one.

## Points, prizes, and what comes after

### The scoreboard

Maximum score is 35: two 1 pointers, two 3 pointers, two 5 pointers, one 7 and one 10. Students run the scoreboard cell at each checkpoint and report their total. Because points come from finished challenges, the score sheet is also your judging sheet. No panel, no deliberation, no arguments.

- If running competitive streams: junior (grades 6 to 9) and senior (grades 10 to 12), top score in each wins. This is exactly how Callysto ran theirs, with 50 dollar gift cards.
- If running non competitive: everyone who delivers a pitch in the sharing circle "gets hired" and the score is personal progress.

### The career thread

Say these out loud during the day, they are the reason the theme exists.

- Challenge 2 sorting: "Every sports network does exactly this to make their top ten lists."

- Challenge 3 filtering: "This is a scout's actual Tuesday: narrow two hundred players to five under budget."
- Challenge 4 charting: "Data analysts at pro teams present exactly this chart in the front office."
- Challenge 6 pitch: "Sports analyst, business analyst, data scientist. Different rooms, same sentence: here is my recommendation, here is my evidence."

## **Adapting this to other careers**

The structure is a template. Swap the dataset and the story, keep the seven challenges identical. Planned variants: music industry analyst deciding which artist a label should promote, wildlife scientist deciding which habitat gets protected. The build guide covers cloning a new track step by step.

## **Credits**

Challenge structure adapted from the Callysto hackathon mini challenge templates (callysto.ca, Creative Commons Attribution license), rebuilt by MindFuel with a new answer checking approach designed for students with no coding background.